特集「バッテリー循環システム」

［解説］

Towards Circularity in Lithium-ion Battery Value Chains: Visions of a Repurposing Ecosystem for Electric Vehicle Batteries

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Synopsis: With the increasing adoption of electric vehicles (EVs), a surge in first end-of-life (FEoL) lithium-ion batteries (LiBs) is expected to become available for either treatment or further use. Since recycling processes are currently insufficient to meet treatment demands in a sufficiently sustainable way, repurposing LiBs for less demanding applications, such as household energy storage systems (ESS), may offer significant economic and environmental benefits. This paper explores the potential for establishing circular value chains for LiBs, focusing on the ecosystem required for their repurposing. Key stakeholders in the value chain are identified, and examples from retail and original equipment manufacturer (OEM) perspectives are discussed. Barriers and opportunities for further development are briefly discussed, with recommendations for future research to ensure a robust, circular system for LiB repurposing.

Keywords: Repurposing; Remanufacturing; Circular value chains; Circular Business Models; Lithium-ion Batteries

1. Introduction

The global transition to electric mobility has led to a sharp increase in the production and usage of lithium-ion batteries (LiBs), in particular in electric vehicles (EVs) (IEA 2024). While these batteries are essential for reducing greenhouse gas emissions from transportation, their lifecycle presents significant challenges. The first end-of-life (FEoL) LiBs from electric vehicles will become available in vast quantities in the coming years, as evidenced by projections from several major markets, including Germany, Norway, and Japan (Kishita et al. 2024). Current recycling technologies being mature enough for large scale implementations, are unable to recover all critical materials effectively (Börner et al. 2022; Patel et al. 2024). This has implications for sustainability, as an increasing demand of battery capacity in both mobility and storage applications would potentially lead to resource scarcity and environmental degradation. Nevertheless, recycling would be inevitable in order to keep materials and resources in the loop (Patel et al. 2024). One promising alternative to recycling FEoL batteries is repurposing for less demanding applications. This approach extends battery lifespans, reduces waste streams and can delay recycling until technologies for recycling Li-ion batteries further improve (Wrålsen et al. 2021). An FEoL battery which still has 80% of its initial capacity left, has potential for further utilisation before it must be discarded, typically at a state of health of 60% (Casals et al. 2019). Among various second-life applications, stationary energy storage systems (ESSs) for households have gained increased attention in literature (Heymans et al. 2014; Thakur et al. 2022; Philippot et al. 2022). ESSs enable the storage and use of renewable energy, such as solar power, and offer households the ability to become "prosumers"—both producers and consumers of energy, including benefits of peak shaving and load levelling (Thakur et al. 2022; Pantelatos et al. 2025). Such applications align with global energy goals and represent a potentially huge market (Hossain et al. 2019).

However, realizing an upscaling of repurposing FEoL LiBs for such lower demanding applications, requires well-functioning collaboration between different actors in the EV battery value chain. These range from EV manufacturers and repurposers to retailers and end-users, but in an emerging value chain the reciprocal links with policy and regulation development should also be taken into account (Chirumalla et al. 2024). This paper explores some of the systemic challenges and

opportunities in building such a system of collaborating stakeholders by presenting and discussing the role of key stakeholders, and by presenting speculative examples of projects where key aspects of stakeholder collaboration, and in particular user perspectives, were explored. This article is based on research and design projects done in the context of the HolE-LiB project (Developing a Holistic Ecosystem for Sustainable Repurposing and/or Recycling of Lithium-ion Batteries).

2. Methodology

This paper uses findings from different sources, including earlier published literature reviews (Ahmed et al. 2023; Pantelatos et al. 2023) and interviews with key stakeholders relevant for future battery value chains in Norway (Ahmed et al. 2025; Pantelatos et al. 2025). Furthermore, a document analysis of media articles and transcribed podcasts was conducted to supplement the literature review and interview data (Pantelatos et al. 2025), and preliminary insights from a yet unpublished Delphi Study on the future of repurposed battery application have also contributed to views presented in this paper. Finally, examples were taken from speculative design projects that focused on exploring aspects of future EV-LiB repurposing value chains. Speculative scenarios through visualizations of potential collaboration between stakeholders and interaction with end users are in design research considered as meaningful research exercises to explore innovative solutions and uncover hidden opportunities within complex systems (Dunne and Raby 2024). By imagining plausible scenarios, researchers can identify synergies, gaps, and tensions between stakeholders that may not be immediately apparent in the current state of the value chain.

3. Stakeholders in a Circular Li-ion Battery Value Chain

The combined insights from the methods deployed in our research so far, and as reported in earlier publications from our research project, suggest that the development of a robust value chain for repurposing FEoL LiBs relies on the active participation of diverse stakeholders. Different constellations of stakeholders, or ecosystems of sustainable business models can be observed to emerge. Depending on the constellation, each stakeholder will need to contribute with specific expertise, resources, and decision-making authority critical for overcoming technical, economic, and regulatory challenges. This chapter explores the roles and significance of key stakeholders, as identified in the literature, and elaborates on their contributions to creating a circular ecosystem.

3.1 Original Equipment Manufacturers (OEMs)

OEMs are pivotal stakeholders because they design and manufacture EV batteries, which directly influences their potential for repurposing. They have the responsibility to design batteries with end-of-life (EoL) considerations in mind, which may include using modular designs and standardizing components, thereby simplifying the process of disassembling and repurposing batteries (Hu et al. 2022; Börner et al. 2022). This ensures that the economic and environmental costs of transitioning batteries to second-life applications are minimized. However, a plethora of competing design criteria challenges circular design approaches (Börner et al. 2022). This could be in terms of production cost, product performance, legal aspects etc. For instance, the adoption of Lithium-Iron-Phosphate batteries, characterized by their lower cost and power density, has led to the development of cell to pack technologies. In these configurations, cells are glued together in order to optimize the power to weight ratio of the battery pack. However, this approach can significantly impede the potential for disassembly and repairability (Scott et al. 2023).

Responsibility for the entire second-life value chain will require huge investments, and therefore collaborations with companies dedicated to repurposing batteries (Section 3.2) as well as retailers and distributors of products with repurposed batteries (Section 3.3) will allow OEMs to contribute to a consistent transition of batteries from EVs to other applications. Such partnerships can also provide feedback on how to improve battery designs for greater efficiency in second-life applications.

3.2 Battery Repurposers

Battery repurposers are stakeholders that evaluate, refurbish, and adapt FEoL batteries for second-life use.

They use their technical expertise in assessing the state of health (SoH) of FEoL batteries, to determine their viability for less-demanding, second-life applications such as household ESSs. This repurposing process may include dismantling the battery pack to module and cell level, characterizing and grading the cells and integrating new battery management systems (BMS), safety features, or casings to meet regulatory and market requirements (Lee et al. 2021). The process requires advanced diagnostics and technical know-how. Ownership of such knowledge can be considered not only as a requirement to survive in the market, but therefore also as a barrier for streamlining value chains towards upscaling applications based on repurposed batteries. Some battery repurposers may also outsource this need or directly acquire pre-checked batteries suitable for second life applications from other stakeholders.

3.3 Retailers and Distributors

In cases where OEMs or repurposing companies will not sell directly to end users, retailers and distributors will likely play a crucial role in communicating the value of products based on repurposed batteries to end-users. Offering such products alongside related products such as solar panels and other ESSs will make it easier for consumers to adopt circular energy systems. They may also play crucial roles in the education of consumers by bearing the responsibility of informing them about the benefits of second-life products, addressing potential concerns about safety, performance, and longevity. Retailers may also further contribute to lowering barriers to adoption by adding value through offering product-service systems (PSSs) by bundling products with installation, maintenance, or financing services. On the other hand, retailers, including dealerships that sell OEM cars, might also play an important role in collecting FEoL batteries from end-users to be repurposed. However, research shows that car dealers still lack the EV LiB circularity knowledge (Bonsu 2020).

3.4 Regulatory Bodies and Policymakers

In addition to OEMs, repurposers, retailers and distributors, government agencies and regulatory bodies are to be seen as essential for creating an enabling environment for circular battery ecosystems where end-of-life Li-ion batteries can be safely repurposed (Wrålsen et al. 2021). Our data suggests that there is a strong need for regulatory bodies to clearly define safety and performance standards for repurposed batteries, ensuring consumer protection and fostering trust in the market. These standards need to address concerns specific to second-life applications, such as explosion and fire hazards and remaining lifetime. Especially in terms of safety, the necessity for mandatory qualifications and certifications for individuals handling repurposed LiBs is outlined in our data. Policymakers can also have a role in incentivizing circularity of batteries by including proper handling of FEol batteries in Extended Producer Responsibility (EPR) schemes, and by offering subsidies, tax incentives, or other financial mechanisms that reward companies for extending battery life (Quinteros-Condoretty et al. 2025). Such incentives may lower economic barriers for stakeholders and make circular business models more viable. However, too many regulations may also hinder the adoption of the second-life batteries (Bonsu 2020).

3.5 Energy Providers

Energy providers may also play an important role in repurposing FEoL batteries into solutions such as decentralized ESSs. A key responsibility would be facilitating grid integration whereby (renewable) energy providers can leverage repurposed batteries as energy storage solutions for households with solar panels or wind turbines, enhancing grid stability and energy independence. These batteries can store excess energy during production peaks and release it when demand is high. In a well-functioning value chain, one can imagine collaborative business models where providers work together with repurposers to bundle second-life ESS solutions with renewable energy systems, creating synergies that make renewable energy systems more attractive to consumers while extending the market for repurposed batteries.

3.6 Consumers and Households

Finally, consumers - whether households or for instance small enterprises - will be vital to the success of circular battery ecosystems. As the end-users who purchase products and PSSs incorporating

repurposed batteries, such as household ESSs or mobile power units, their participation is vital. This market demand will eventually create the economic incentive for stakeholders to invest in circular battery solutions. Adoption success will depend on aspects like consumer trust in the safety and reliability of second-life batteries, and lack of attention to this may significantly hamper market growth. Therefore, safety considerations during the repurposing process such as assessing the correct SOH of cells and monitoring the cell temperature through BMS during operation is necessary to ensure high safety level (Lee et al. 2021). Awareness campaigns and transparent product information will be essential to address misconceptions about second-life products. Consumers will also provide valuable feedback on the performance of second-life products and concerns they may have, which will help repurposers and retailers to improve the products and PSSs that they offer. Figure 1 provides a simplified visualisation of important stakeholders in future repurposed battery value chains, and examples of how they connect. The numbers in the figure correspond with speculative examples in connection to which stakeholders are involved, as presented in Chapter 5.

4. Emerging repurposed EV battery value chains in different countries

(Kishita et al. 2024) present several examples of market initiatives towards a more systematic exploitation of FEoL batteries, together with a comparison of existing background context around this practice in Japan, Germany and Norway. In Norway, several small and medium-sized companies provide an initial range of products for both industrial, residential and power grid contexts. Some companies focus heavily on technology development and collaborate with OEMs (Ahmed et al. 2025) such as ECO STOR that collaborates with Nissan and has developed a reconfiguration interface that allows to reprogram the original BMS and reuse the original battery pack as is. The company Hagal has developed a technology to monitor and control each individual cell that makes the use of a central inverter unnecessary and thereby reduces losses. Market examples are still few but include some initiatives in the commercial and public sector, such as the company Alternativ Energi, which provides energy storage products based on repurposed LiBs for the Norwegian cabin market (Ahmed et al. 2025). In Germany, different constellations of OEMs have established large ESSs for the German power grid at different locations. Also here,

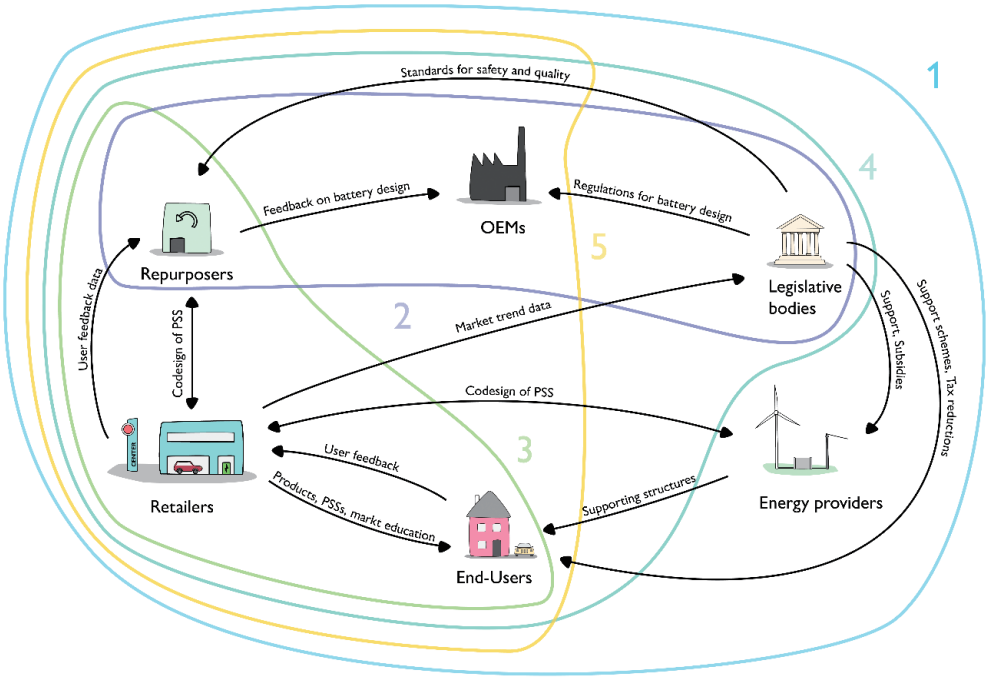


Figure 1: Important stakeholders in future repurposed battery value chains

several start-ups have emerged like Betteries who offer mobile energy storage solutions based on modular storage blocks from repurposed EV LiBs sourced from Renault group through their brand Mobilize. From Japan, a prominent collaboration between different stakeholders in the context of repurposing used batteries from EVs is 4R Energy, which is a company jointly founded by Nissan Motor Corporation and Sumitomo Corporation. The company sources batteries from Nissan Leaf cars that have reached their end of life, and sorts modules into various grades depending on their state of health. The best, A-grade modules, are used for refurbishing Nissan Leaf battery pack, whereas B-grade modules are sold to other companies and repurposed into a variety of applications, including industrial energy storage, powering railway level crossings or portable energy storage for residential use such as by Nissan and JVC Kenwood.

5. Speculative exploration of elements of well-functioning Value Chains for Second-Life Batteries

As argued for in the previous chapter, future value chains for second-life batteries will be inherently interconnected, regardless of whether the future will see one dominant version or a variety of value chains. For example, OEMs rely on feedback from repurposers and consumers to improve battery designs, while regulatory bodies must align safety standards with the technical capabilities of repurposers. Similarly, retailers depend on effective partnerships with repurposers and energy providers to deliver value-added solutions to households. Recognizing these interdependencies is critical for fostering collaboration and ensuring that all stakeholders contribute effectively to a sustainable circular ecosystem.

In connection to the HolE-LiB project, master students were challenged to develop speculative designs with the aim to explore, analyse and visualise how collaboration between various stakeholders could materialise. This chapter draws on speculative examples drawn from both final and pre-master thesis projects, as well as interdisciplinary student projects, which illustrate how different scenarios can operationalize circular battery systems from the perspective of retailers, OEMs, consumers and end users. The examples mostly draw on exploring end-user perspectives, because this aspect is least addressed in literature so far (Pantelatos et al. 2023).

5.1 Speculative example 1: Exploring business models and value chains

The first example (number 1 in Figure 1) is taken from the interdisciplinary course Experts in Teamwork (EiT) at NTNU. The course ran in Spring 2023 and focused on the theme 'Building sustainable solutions for the Lithium-ion value chain'. Master students from diverse educational backgrounds worked in five group projects focusing on different aspects of the battery value

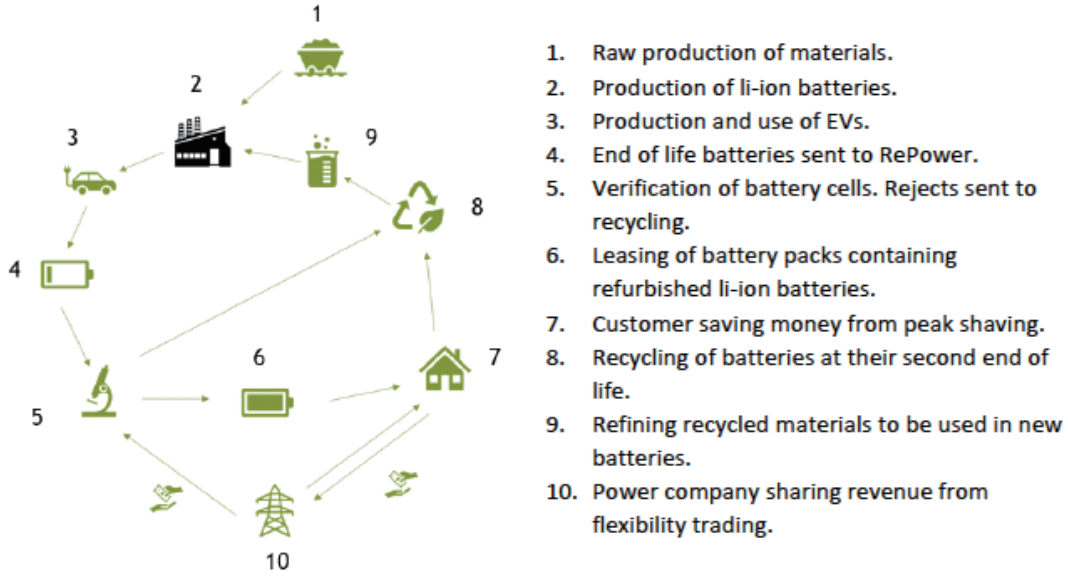


Figure 2: Speculative business model RePower

chain such as business models, collection of the EoL batteries, policies and regulations, and recycling. One of the projects explored the potential for integrating refurbished LiBs into household ESSs in Norway. It examined factors influencing user acceptance, identifies barriers and drivers for second-life battery adoption, and proposes sustainable business models under a circular economy framework. Key insights were gathered through industry interviews and literature reviews. The proposed "RePower" business model (Figure 2) emphasizes leasing refurbished batteries for energy storage, enabling cost savings through peak shaving and grid flexibility. The refurbished batteries would have an automatic monitoring system that will charge when the electricity prices are low and discharge for use when the prices are high. The business model depends on collaboration across different stakeholders in the battery value chain with an emphasis on a contractual partnership with EV manufacturers providing the EoL batteries. Moreover, 'RePower' also considers energy flexibility trading, that can be beneficial to energy trading companies who earn revenue through energy trading and require increased power storage. Figure 2 shows how value is generated by RePower during the flow batteries from production to recycling.

5.2 Speculative example 2: Mapping regulatory complexity

A second Experts in Teamwork project (number 2 in Figure 1) focused on the collection and transportation of FEoL batteries. It identified barriers such as financial constraints, regulatory gaps, safety concerns, variable battery designs, and limited infrastructure for reverse logistics. Interviews with companies like Batteriretur and NorSirk provided insights into existing collection practices and safety measures. One of the main findings from the project emphasizes the complexity of the regulations related to the collection, transportation and how it increases overall cost of repurposing FEoL LiBs as they fall under the hazardous goods category under the UN dangerous good list. This entails that the EoL battery must follow certain special provisions for packing and transporting. For example, special provision (SP) 188 has one of five requirements that the EoL battery pack must have a watt-hour rating for a LiB cell at 20 Wh. Figure 3 shows what a regulatory compliance table could look like for an EoL battery pack.

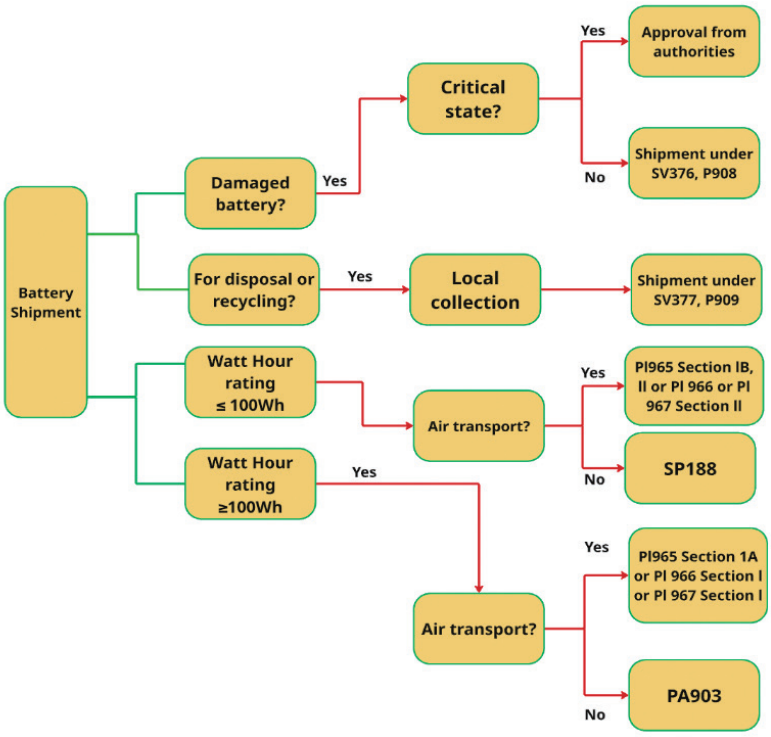


Figure 3: Complexity of regulations related to SOH and collection practices for battery repurposing

5.3 Speculative example 3: Exploring user-friendliness and retail integration

The third example (number 3 in Figure 1) is taken from a master project in Industrial Design Engineering, which aimed to create speculative scenarios for collaboration between battery repurposers and retailers (Jaimes 2023). One scenario imagines a fictional remanufacturing company named Norsk-e which offers used battery modules to consumers who do not require or cannot afford an entire battery pack. These modules can be purchased and transported as needed or connected to a grid for charging. The modular design allows for flexibility of stacking as many modules as necessary to create a custom battery back that suits one's needs and financial resources. To illustrate this scenario, a speculative advertisement was created, linking the imaginary Norsk-e product to an existing well-known electronics retail company (Figure 4). Similarly, another speculative scenario concerned the offering of a refurbised battery back from an EV with 55 kWh capacity, the capacity to power a cabin with essential electrical applications unterinterrupted for 10 days without charging – a product ideal for Norwegian leisure context. The battery pack design includes three modules which can be easily detached and used for other activities, by using a conditioned backpack with an inbuilt energy converter suitable for use during outdoor activities such as hiking, camping, canoe trips or other activities in remote areas that require mobile electric power.

5.4 Speculative example 4: consumer information and education

Another speculative project (number 4 in Figure 1) was grounded in both secondary and primary user insights - the latter in the form of surveys and workshops - which led to the understanding that the successful use of repurposed LiBs from EVs into consumer products would require several conditions to be met (Hjelm-Hansen and Skogseth 2024). In particular, concerns related to capacity and safety would need to be addressed, but users would also need confirmation that the product they would buy would be of high quality, indeed contribute to sustainability, and indeed be based on repurposed batteries. The students decided to test what a hypothetical label for repurposed batteries could look like. The label should be based on the UL 1974 standard, which demonstrates quality, safety and ethical production of recycled batteries. The goal of the project was to investigate potential label design, specifically targeted at users, by being easily visible and recognizable. Based on the user research, five different personas were created, each with their own different main concerns. Participants of the study had to state which persona they identified most with, and were presented a set of labels which they would place to the factor they thought would give the best association. Figure 5 shows the result of the experiment, where each cell depicts the top three labels preferred by the group of respondents who identify with a certain persona (column on the left), for each factor. The other columns show the respondents' answers according to each factor. The diagonal



Figure 4: Speculative design project Norsk-e

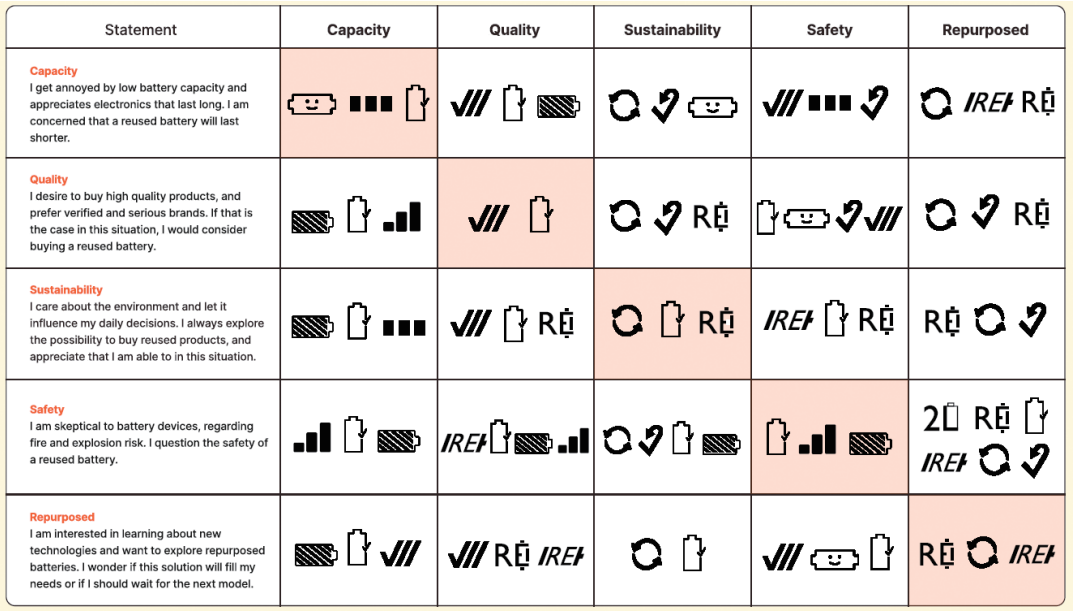


Figure 5: Labels describing certain factors sorted through different respondent groups

highlights what the group answered in the category they identified most with.

5.5 Speculative example 5: building consumer trust

From a retail perspective, value chains will need to focus on bringing second-life batteries to markets in a way that aligns with consumer needs and market dynamics. Retailers serve as intermediaries between repurposers and end-users, offering products that meet functional, economic, and sustainability criteria. This fifth example (number 5 in Figure 1) presents a speculative brand Rehohm which repurposes electric car batteries into household ESSs (Kaarud 2024). The project had a particular focus on end-user perspectives relevant to user acceptance, and included development of a brand identity, a product design language, and website mock-ups for a fictional supplier. Creating familiarity with the product in a known context was explored as a strategy to mitigate safety concerns. The project builds on data collected through an online survey, expert interviews, workshops with homeowners as potential users of repurposed ESSs and graffiti walls at public spaces where people were encouraged to write their first association to batteries. The collected



Figure 6: Speculative design project Rehohm

data was synthesized using affinity mapping, user personas, and empathy maps to compile insights, revealing diverse sets of attitudes, associations, and beliefs related to batteries in general and repurposed batteries more specifically. The initial research of the project pointed out that some perceptions were quite contradictory, making it complex to address all views in product communication and design. For instance, emphasizing fire safety might introduce concerns to those who initially had none. Thus, discussing battery safety can increase worries for some while reassuring others, and efforts to assure consumers of a company´s reliability might inadvertently create scepticism. Furthermore, the focus on engineering specifics, such as the level of disassembly in the repurposing process, may be too technical for some, but essential for others. The dilemma of battery capacity size highlights conflicting interests between safety and economic gains: larger capacities pose greater perceived risks, but also higher economic gains. Additionally, smaller ESSs may be seen as "useless", while larger ones could be aesthetically displeasing. A rigid, tough design might enhance safety perceptions but compromise aesthetics. Furthermore, sustainability efforts can be paradoxically perceived as greenwashing by some. In the final design, the fictional supplier's brand identity was designed to convey the perception that repurposed electric vehicle batteries are environmentally friendly, and that the product is safe and of high quality through association with high-quality car brands from which the repurposed batteries were taken. This was done by using the aesthetic expression of the grille of the car as basis for the exterior design of the ESS, thus creating such associations with the car brand.

6. Discussion and conclusion

The speculative examples provided here highlight the potential for innovations in value chains and solutions, but they also underscore significant challenges. In view of the survey results, media sources and interviews with stakeholders and prospective users done in the context of the mentioned examples as well as our prior research, it can be confirmed that safety concerns are of utmost importance to address. Rational safety concerns will need to be mitigated through rigorous testing and certification to ensure safety in household applications. Regulatory bodies will play an essential role to ensure these will be in place, but may be challenging to get into place, given the global nature of the topic and different priorities given to topics such as resource availability, sustainability goals and customer safety. Purely emotional or even irrational safety concerns associated with batteries may take various forms; for some end-users the use of repurposed batteries will bring associations of unsafety no matter how these are rationally addressed. For others, repurposed batteries may be considered as 'proven to be safe', whilst a third group may have no concerns at all and may not become aware about safety as an issue at all until it is explicitly mentioned in consumer information. Further research on the topic is needed, both academically but also by value chain stakeholders themselves, pertaining to specific geographic and value proposition contexts.

Secondly, cost is a factor of major significance. Prices of new LiBs are decreasing both because of advancements in battery technology and a volatile market. In turn this puts a strain on the feasibility of repurposing LiBs, where second life products logically would compete with new ESS solutions. The safety concerns mentioned above are related to the hazardous nature of LiBs which potentially result in thermal runways (Chen et al. 2021). As the risk of failure of second-life LiBs is related to the number and characteristics of charge and discharge cycles, thorough testing and certification are essential to ensure their safety in household applications, and this safety assurance can also lead to higher costs. Expertise in determining state-of-health of used batteries is likely to be a crucial asset for becoming successful on the market, and cooperation with OEMs who are in possession of user data from the first use cycle will in many cases be a huge advantage, as shown for example in 4R Energy's practice.

Furthermore, supply chain coordination remains a challenge. Effective collaboration across multiple stakeholders, including EV manufacturers, battery repurposers, and retailers is required to establish reliable supply chains. Achieving favourable cost of using repurposed batteries is likely to require economies of scale in terms of acquiring and processing of user batteries, which may hamper the success of small entrepreneurial firms hoping to get a slice of the market. The speculative examples illustrate how

innovative approaches can help overcome existing gaps and unlock the potential of second-life batteries for household applications. By integrating design for circularity, modularity, and community-focused solutions, value chains can become more robust and sustainable.

In sum, the circularity of LiBs for EVs represents a critical challenge and opportunity in the transition to sustainable energy systems. Repurposing FEoL batteries in applications such as household ESSs may significantly contribute to this goal, but only through the collaboration of diverse stakeholders and the development of robust value chains. This paper provides some speculative scenarios that illustrate how such collaboration could materialise, pointing out a number of challenges pertaining to legal, economic, technical and social barriers, to be addressed.

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